

Cooper Wooley

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Education

Missouri University of Science and Technology

B.S. Computer Science, Mathematics Minor | GPA 3.84

Dec 2025

Relevant Coursework: ML, Deep Learning, LLMs, Algorithms, Data Structures, Databases, OS, Networks, Security

Technical Skills

- **Languages:** Python, C/C++, JavaScript, SQL, HTML/CSS
- **Tools:** PyTorch, Linux, Docker, Git, VS Code, Arduino IDE, LaTeX

Experience

Formula Electric Design Team

Software Member

Aug 2023 - Present

- Designed and implemented embedded software for microcontrollers (Teensy, ESP32) using C++ in the Arduino IDE to support the team's electric race car systems
- Extended dashboard class architecture to add object-oriented display control for dual-screen setups, implementing static/dynamic text rendering with hash maps for efficient updates
- Developed an interactive data acquisition (DAQ) visualization interface using JavaScript and Chart.js for real-time telemetry and sensor analytics
- Collaborated to board state logic, auxiliary function libraries, and early sensor testing frameworks to accelerate development for new team members

City of Sikeston - Public Works, Parks and Recreation

Sikeston, MO

Seasonal Skilled Worker

May 2023 - Aug 2025 (Summers)

- Led field maintenance for baseball and soccer setups, ensuring optimal event preparation
- Coordinated daily field marking and equipment setup
- Participated in tornado and flood recovery efforts, safely clearing debris and restoring public spaces

Sikeston Public School District

Sikeston, MO

IT Internship

May 2022 – Aug 2022

- Established internal network infrastructure by configuring distribution frames
- Deployed switches and wireless access points; terminated fiber optic and Cat6 cabling
- Performed hardware troubleshooting and system reimaging

Projects

ParkVision2.0 Real-Time Paring Analytics System - (Python, YOLO, DeepSORT, FastAPI, OpenCV, Docker)

- Transformed static parking detection prototype into full-featured intelligent parking analytics system providing real-time occupancy tracking, license plate recognition, and interactive lot management tools
- Developed end-to-end computer vision pipeline using YOLO (Ultralytics) and DeepSORT for real-time vehicle detection and multi-object tracking, implementing custom polygon-to-bounding-box IoU matching algorithms for accurate parking lot occupancy analysis
- Built session management system tracking vehicle entry/exit events and parking durations, with timeout-based logic for handling vehicle disappearance and generating completed session events
- Integrated computer vision module with FastAPI backend through REST API endpoints, enabling real-time updates via callback functions and batch processing, and containerized application using Docker with GPU support for CUDA-accelerated inference

Copyright Benchmark for Image Generation Models - (Python, PyTorch, CUDA, Transformers, Computer Vision)

- Built automated image generation pipelines for OmniGen2 and DALL-E Mini models using PyTorch, implementing GPU memory optimization and custom scheduler configurations to process batch image generation from JSONL prompt files
- Developed production-ready scripts with virtual environment automation, error handling, and resource management for generating copyright test cases used in comprehensive AI model evaluation benchmarks